

Amazon VPC (Virtual Private Cloud) - Service Overview

Introduction

Amazon Virtual Private Cloud (VPC) lets you provision a logically isolated section of the AWS cloud where you can launch AWS resources in a virtual network that you define. You have complete control over your virtual networking environment, including IP address ranges, subnets, route tables, and network gateways.

CIDR Blocks and Subnets

When you create a VPC, you specify an IPv4 CIDR block (for example, 10.0.0.0/16), which defines the range of private IP addresses available within that VPC. The VPC is then divided into subnets, each associated with a specific Availability Zone and a subset of the VPC's CIDR range.

Public vs Private Subnets

- A public subnet has a route table entry directing internet-bound traffic (0.0.0.0/0) to an Internet Gateway, making resources within it directly reachable from and able to reach the internet (if they also have a public IP).
- A private subnet has no direct route to an Internet Gateway. Resources in a private subnet cannot be reached directly from the internet, and can only reach the internet indirectly, typically via a NAT Gateway placed in a public subnet.

Internet Gateway

An Internet Gateway (IGW) is a horizontally scaled, redundant VPC component that allows communication between resources in a VPC and the internet. Exactly one IGW can be attached to a VPC at a time.

NAT Gateway

A NAT (Network Address Translation) Gateway allows instances in a private subnet to initiate outbound connections to the internet (for example, to download software patches) while preventing the internet from initiating connections to those instances. A NAT Gateway requires an Elastic IP address and is deployed within a specific public subnet.

Route Tables

A route table contains a set of rules, called routes, that determine where network traffic from a subnet or gateway is directed. Every subnet must be associated with exactly one route table, though a single route table can be associated with multiple subnets.

Security Groups vs Network ACLs

- Security Groups operate at the instance (or ENI) level and are stateful: return traffic is automatically allowed regardless of outbound rules.
- Network ACLs operate at the subnet level and are stateless: both inbound and outbound rules must be explicitly configured, since return traffic is not automatically permitted.

VPC Endpoints

A VPC Endpoint enables private connectivity between a VPC and supported AWS services (such as S3 or DynamoDB) without requiring an Internet Gateway, NAT device, or public IP address, keeping traffic entirely within the AWS network backbone.

Multi-AZ Design

Production VPC architectures typically span at least two Availability Zones, with a public and private subnet pair in each AZ, so that the failure of a single AZ does not take down the entire application.

Worked Subnetting Example

Given a VPC with CIDR block 10.0.0.0/16 (providing 65,536 addresses), a common pattern across two Availability Zones is: public subnet A at 10.0.0.0/24, public subnet B at 10.0.1.0/24, private subnet A at 10.0.10.0/24, and private subnet B at 10.0.11.0/24, each /24 providing 256 addresses (251 usable, since AWS reserves 5 addresses per subnet for internal networking purposes). An RDS database and the ECS tasks running application logic would typically live in the two private subnets, while only the load balancer and NAT Gateways sit in the public subnets.

VPC Peering and Transit Gateway

VPC Peering creates a direct, private network connection between two VPCs, allowing resources in each to communicate as if they were on the same network, but peering connections are not transitive: if VPC A is peered with VPC B, and VPC B is peered with VPC C, A cannot reach C through B. AWS Transit Gateway solves this at scale, acting as a central hub that can connect many VPCs and on-premises networks together, avoiding the need for a full mesh of individual peering connections as the number of VPCs grows.

Common Use Cases

- Hosting a multi-tier web application with a public-facing load balancer and private application/database subnets
- Isolating workloads between environments (dev/staging/prod) using separate VPCs

- Establishing private, secure connectivity to on-premises networks via VPN or AWS Direct Connect

Best Practices

- Place compute resources like ECS tasks or EC2 instances running application code in private subnets, never directly in a public subnet unless a public IP is strictly required.
- Use one NAT Gateway per Availability Zone in production to avoid cross-AZ data transfer charges and single points of failure.
- Use VPC Endpoints for AWS services accessed frequently from private subnets, such as S3, to reduce NAT Gateway data processing costs.